Second Opinion

A diagnostic model that reports when it should not be trusted.

UnivaBio 2026 — Artificial Intelligence for Human Health · Research prototype, not a medical device · Python, scikit-learn, Streamlit

THE PROBLEM

Ask almost any health-AI demo for a prediction and it will give you one. Feed it a patient unlike anyone in its training data, a mistyped measurement, or a sample from a different lab, and it *still* gives you one — with the same confident percentage attached. The number carries no signal about whether the model had any business producing it. This is a central reason clinical machine learning struggles to deploy: a model validated at one hospital degrades at the next, and nothing in its output says so.

THE APPROACH — THREE QUESTIONS, NOT ONE

Second Opinion answers three separate questions about every sample, and withholds the diagnosis when the last two say it should.

QUESTION	METHOD	WHY IT IS NOT ENOUGH ALONE
1. What is the probability?	Isotonic-calibrated logistic regression	Always answers, even for nonsense
2. Is that probability trustworthy *for this sample*?	Split conformal prediction (LAC)	Silently voids under distribution shift
3. Is this sample the kind of thing I have seen?	Class-conditional Mahalanobis novelty detection	Says nothing about the diagnosis

Reporting all three — and gating the headline number on the last two — is the product. The model therefore has **four** possible outputs, not two:

```
{malignant} / {benign}   a decisive call
{malignant, benign}   genuinely borderline — a human should look
{}   abstain: nothing is plausible; do not read the percentage
```

That empty set is the point of the project. A model that cannot say "I don't know" will hand a confident answer to a case it has never seen.

KEY FEATURES

- **A diagnosis that disappears.** When novelty or conformal analysis says no, the headline is a refusal rather than a number. The probability stays on screen, explicitly labelled as transparency, not a result.
- **Two withholds, two computed next steps.** A conformal abstain names the smallest single measurement change that would make the call decisive — *"if mean concave points drops 37%, this becomes a 90%-confident benign."* A novelty withhold instead names the measurement that looks like a typo and refuses to speculate past it. Both are computed per sample, not canned text.
- **Catches impossible combinations, not just out-of-range values.** One demo preset sets *mean area* to its true dataset maximum while *radius* and *perimeter* stay at their medians: every value is individually in range, but the geometry is impossible — a nucleus cannot have that area at that radius. Mahalanobis distance flags it because it accounts for how features covary; a per-feature bounds check passes it silently. The base classifier is **100% confident** on this input, and the system reports **no diagnosis**.
- **Exact attribution.** The contribution chart is the model's own arithmetic — coefficient × standardised value, summing exactly to the logit — not a SHAP approximation of it.
- **A stress test shipped in the UI.** Held-out samples are progressively corrupted: the withhold rate climbs to 100% while *confident error* (decisive, in-distribution, and wrong) does not climb with it. It fails by going quiet instead of by guessing.

RESULTS

Wisconsin Diagnostic Breast Cancer: 569 samples, 30 features, 37.3% malignant; three-way train / calibration / test split where calibration is never trained on and test is touched only for these numbers. Averaged over **40 random splits** (mean ± sd): AUC **0.985 ± 0.012**, Brier **0.025**, ECE **0.026** (improved from 0.032 uncalibrated); **conformal coverage 93.5% ± 3.5%** against a 90% target; decisive on **95.1%** of cases, abstaining on **5.0%**. Novelty detection flags **100%** of deliberately scrambled inputs versus 6.8% of genuine held-out samples. Single-split coverage ranges from 86% to 99%, which is why no single split is quoted as the result. Every figure regenerates with `python scripts/evaluate.py` ; **32 claim-level tests** assert each claim made in the README and in the UI.

WHO IT HELPS

Clinicians and triage workflows get a refusal they can act on instead of a confident number they cannot audit. **Lab and QC staff** get a flagged measurement that reads as a transcription check. **Deployment and validation teams** get a built-in monitor for exactly the silent degradation that breaks models at a second site. **Patients** are not assigned a number produced by extrapolation.

HONEST LIMITATIONS — WHAT IS FINISHED AND WHAT IS NOT

This is **not a medical device**: it cannot diagnose anyone, has never been clinically validated, and must not inform a real medical decision. The data is 569 samples from one institution collected in the early 1990s, with no demonstrated transfer anywhere else. It classifies **pre-extracted nuclear measurements, not images** — the hard upstream problem of getting from a slide to those 30 numbers is assumed already solved. The conformal guarantee is *marginal*, not conditional (Angelopoulos & Bates, 2023): it holds across the population, not necessarily within every subgroup, and subgroup-conditional coverage is the obvious next step and **is not implemented**. A per-feature spline GAM was tried and **reverted** after a 25-seed controlled A/B — linear won on all eight metrics (ECE 0.031 vs 0.111) — and is kept in `gam.py` as the record of what was tried.

Methods trace to published sources: Street, Wolberg & Mangasarian (1993, WDBC and upstream feature extraction) · Guo et al. (2017, calibration ≠ accuracy) · Sadinle, Lei & Wasserman (2019, LAC set-valued classifier) · Angelopoulos & Bates (2023, split conformal) · Lee et al. (2018, class-conditional Mahalanobis OOD) · Hastie & Tibshirani (1986, the additive model measured and not adopted). Full strings in `second_opinion/references.py` .

Run it: `pip install -r requirements.txt` then `streamlit run app.py` — select **Implausible geometry** in the cell-profile dropdown, or push any slider past the dataset range, and watch the diagnosis disappear.